

Real-Time Retail Feedback Intelligence

Transforming unstructured customer Reviews of ChicStyle
into actionable strategic insights using GenAI

PETR SPEVAK | MIT AAIDS

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Business Context & Problem definition

Current state

23 486

Customer reviews

Months

Time to detect defects manually

Large number of Returns

due to unnoticed quality issues

Key Pain Points

Scalability

Manual review reading
vs. Sales growth

Reactive vs. Proactive

Critical issues (e.g., faulty zippers) are
found only *after* returns spike

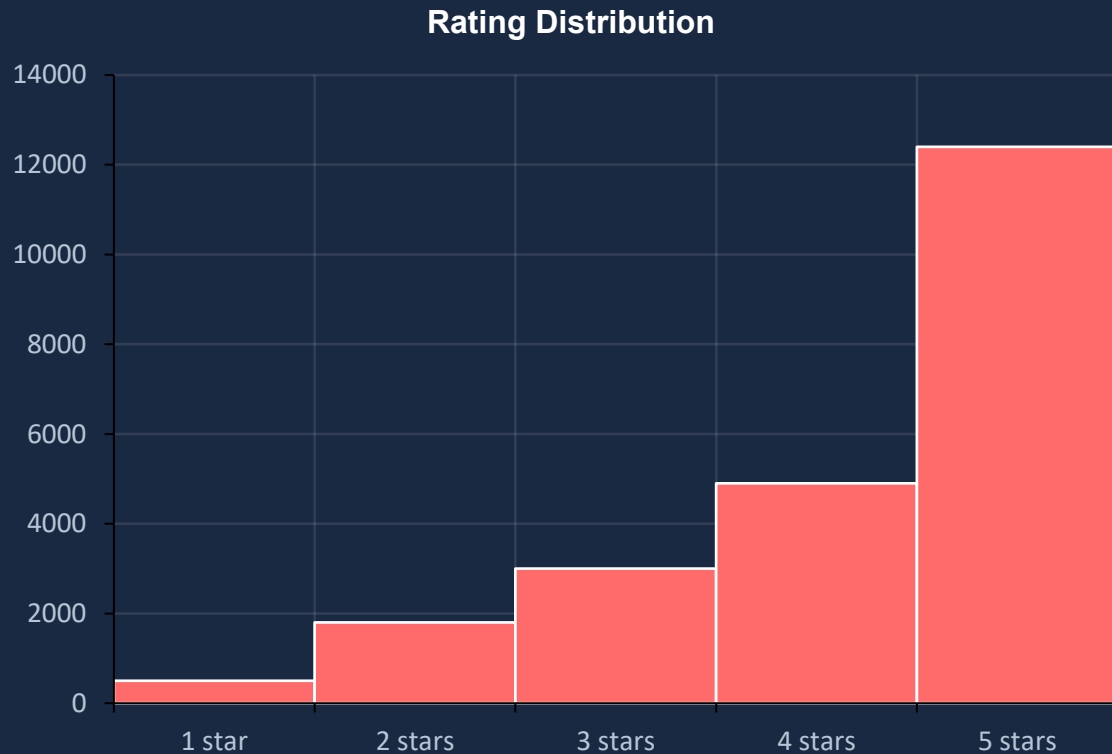
Lost Insights

Valuable Nuances are missed
by basic keyword search

Key Objective:

→ Build an automated GenAI pipeline to parse reviews in real-time
and extract structured data for the QA /Product team

Data & Key Findings (EDA)



(Full Rating Dist. chart in App. A)

Key findings Bullet Points:

★ **Positive Bias:** The majority of ratings are 5-star masking critical negative feedback in the noise

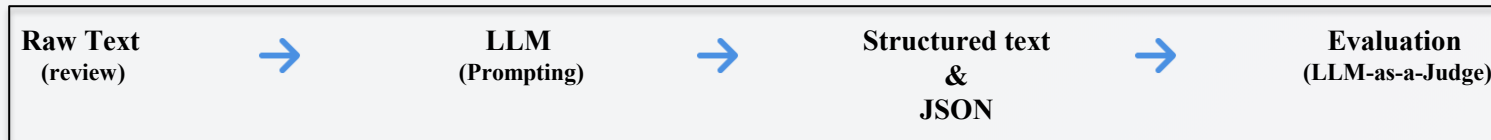
🏷️ **Technical Vocabulary:** Negative reviews (1-2 *) are driven by specific terms like "Fabric", "Material", "fit", "back" .

INSIGHT: Simple sentiment analysis is not enough; we need deep extraction of technical details by AI

(Word-cloud analysis in App. B)

Solution Design | (GenAI Pipeline)

Visual Flow



⚡ Zero-Shot

Direct extraction without examples. Fast baseline

💡 Few-Shot

Providing 3 examples (Pos/Neg/Neutral) to guide structure

🧠 Chain-of-Thought

Forcing the model to reason step-by-step about tone and context before outputting response

Evaluation: Implementet an LLM as a judge system to objectively score extraction quality from 0 to 1

Experimental results | Prompt engineering

🏆 **Chain-of-Thought V1 | GPT-4o-mini**
Is the winner scoring **0,968** extr. Quality by the Judge

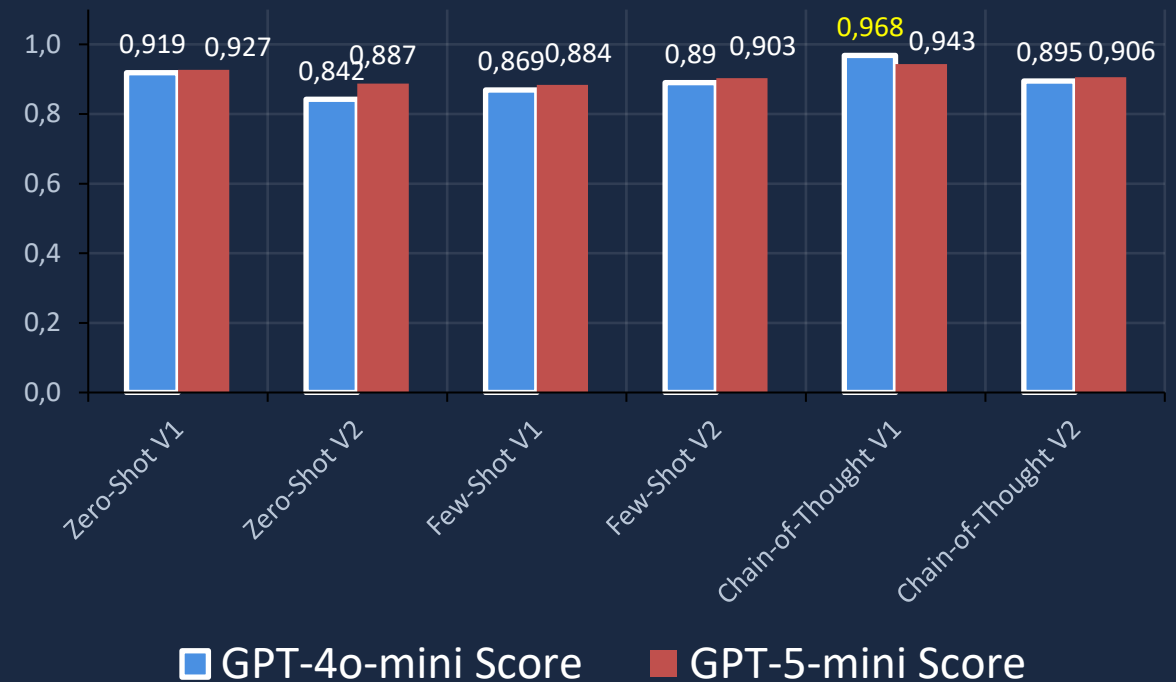
💡 Key Takeaways

→ **Quality over structure** CoT outperformed rigid templates because it allowed the model to analyze nuance (e.g. Great dress, tight armpits)

→ **Complexity Trade-off** Adding too many constraints actually lowered performance compared to allowing the model to think freely

(Detailed comparison of the Prompt Engineering scores in App. C)

Comparing the scores of the techniques



Model Showdown

GPT-4o-mini vs. GPT-5-mini

(Detailed table of the showdown in App. D)

GPT-4o-mini CoT v1

Metric	Value
Accuracy	87%
Precision	1.00
False Positives	0
Speed per Iteration	6.8 sec./it

Conclusion: Lesser Accuracy but with no false positives The perfect Gatekeeper

GPT-5-mini CoT v1

Metric	Value
Accuracy	89%
precision	0.94
False Positives	10
Speed per Iteration	14.43 sec./it

Conslusion: Better overall Accuracy but with many constaraints

Deployment statgegy (The Verdict)

GPT-4o-mini is Fast for production deployment with **0 error rate** by False Positives

Conclusion: Lesser Accuracy but with no false positives means there wont be a mismatch in QA analysis and the dataflow. With double the speed of the GPT-5-mini and lesser usage price it is definetely more scalable for ChicStyle purposes.

Framing risks & constraint of the solution

1 Mixed sentiment complexity

Models Struggle with dual-sentiment reviews (e.g. Love the style, hate the zipper...)

RISK: Misclassification

Possible solution: Human in the loop for low conf. scores

2 Operational Scalability

Benchmark showed that GPT-5 is 2x slower than GPT-4o, there can be issues with cloud hosted LLMs

RISK: Data bottlenecks during sales peaks (e.g. Black Friday) **Possible solution:** Code reinforcement using multiple LLMs as a backup

3 Reliability risks

AI may deal with hallucinations and misinterpret sarcasm or slang as factual product flaws.

RISK: Incorrect data for QA and Product development

Possible solution: Statistical threshold and scheduler model evaluation

Conclusions & Benefits

SCALABILITY

Handling main bussiness event and peaks like Black Friday without adding headcount.

CONSISTENCY

Every review is treated with the same analytical rigor.

TIME SAVINGS

Reduced analysis and sentiment distribution time from Weeks → Minutes.

CLOSING LINE

The solution can move whole organisation from reactive support to proactive product intelligence.

(Detailed comparison of the Sentiment Distribution in App. E) & F)

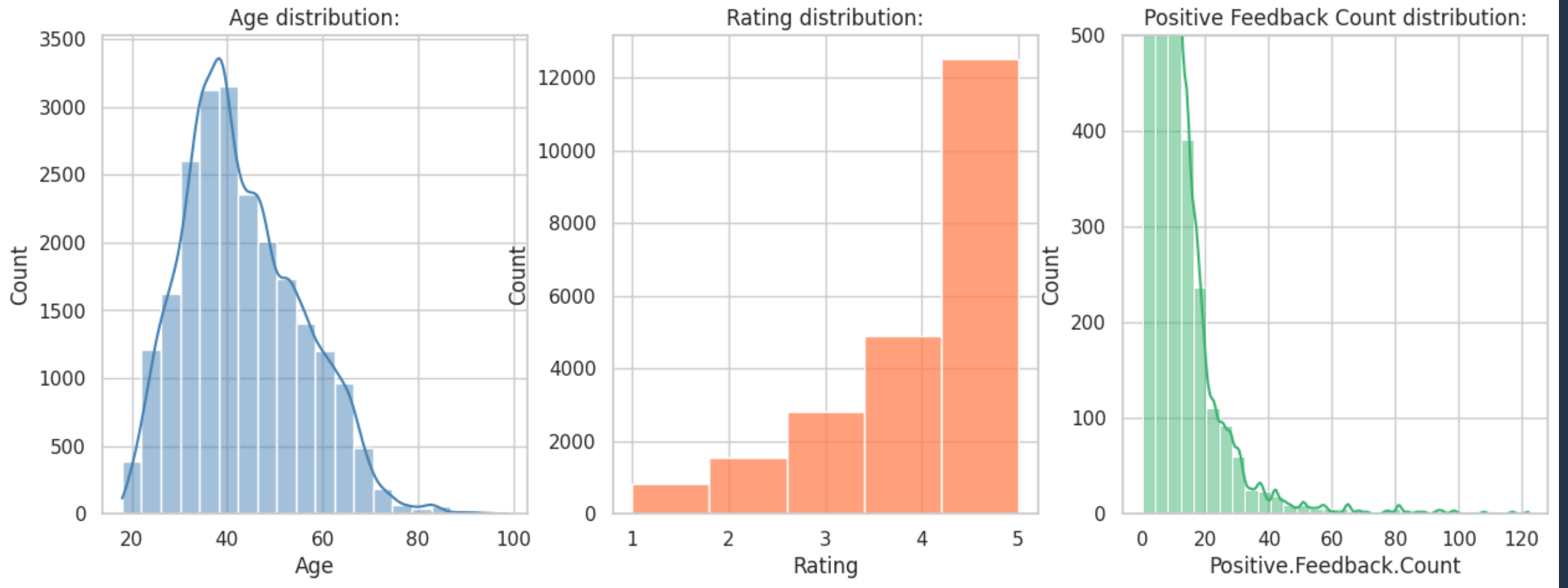
Thank You for your attention!

Open for questions. (Reference Appendix for technical details)

APPENDIX

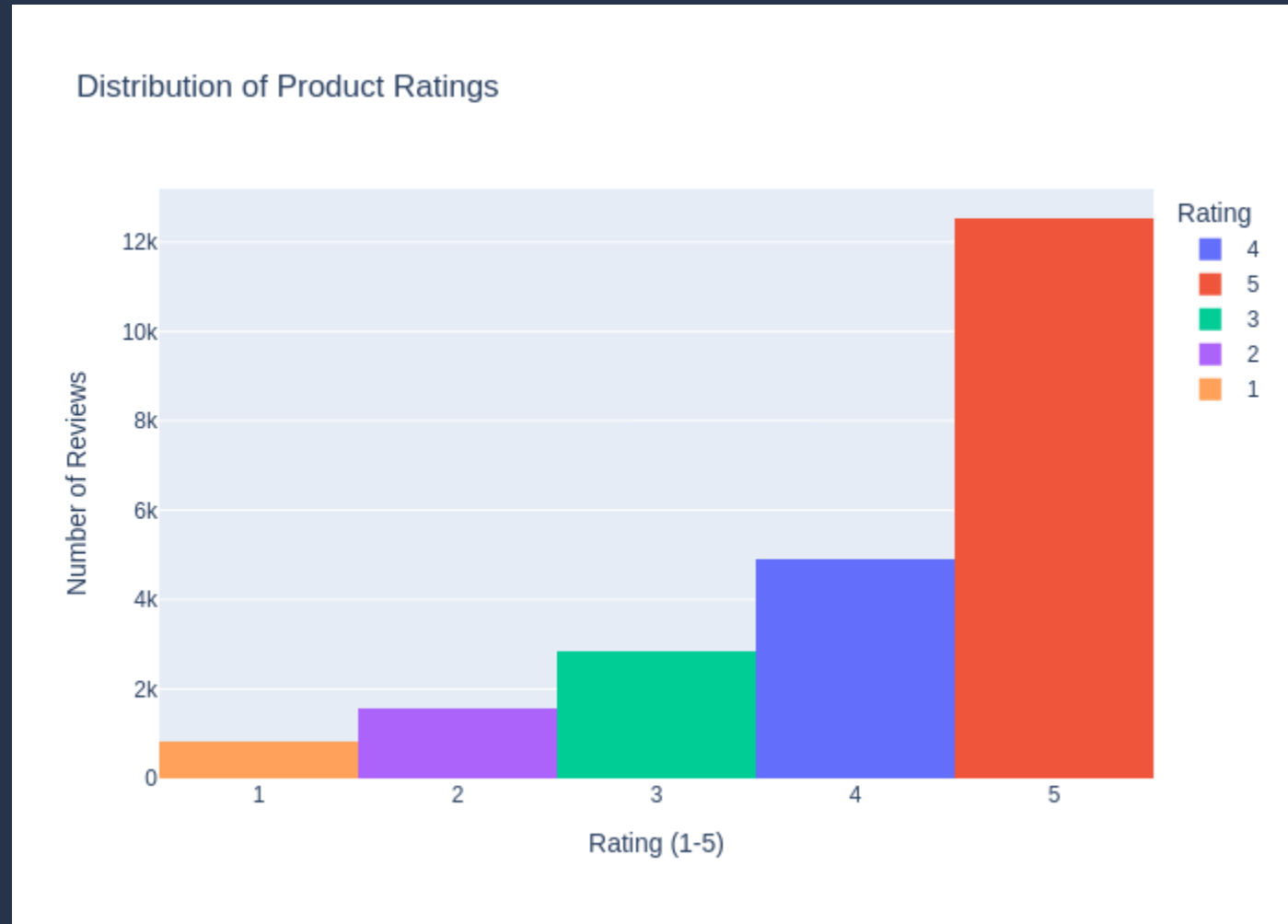
(Further ref. In final research notebook)

A: SUMMARY STATISTICS (EDA)



(Further ref. In final research notebook)

B: Distribution of Product Ratings



(Further ref. In final research notebook)

C: Prompting techniques evaluation table

Prompting Technique	GPT-4o-mini Score	GPT-5-mini Score	Score Delta
Zero-Shot V1	0.919	0.927	+0.008
Zero-Shot V2	0.842	0.887	+0.045
Few-Shot V1	0.869	0.884	+0.015
Few-Shot V2	0.890	0.903	+0.013
Chain-of-Thought V1	0.968	0.943	-0.025
Chain-of-Thought V2	0.895	0.906	+0.011

(Further ref. In final research notebook)

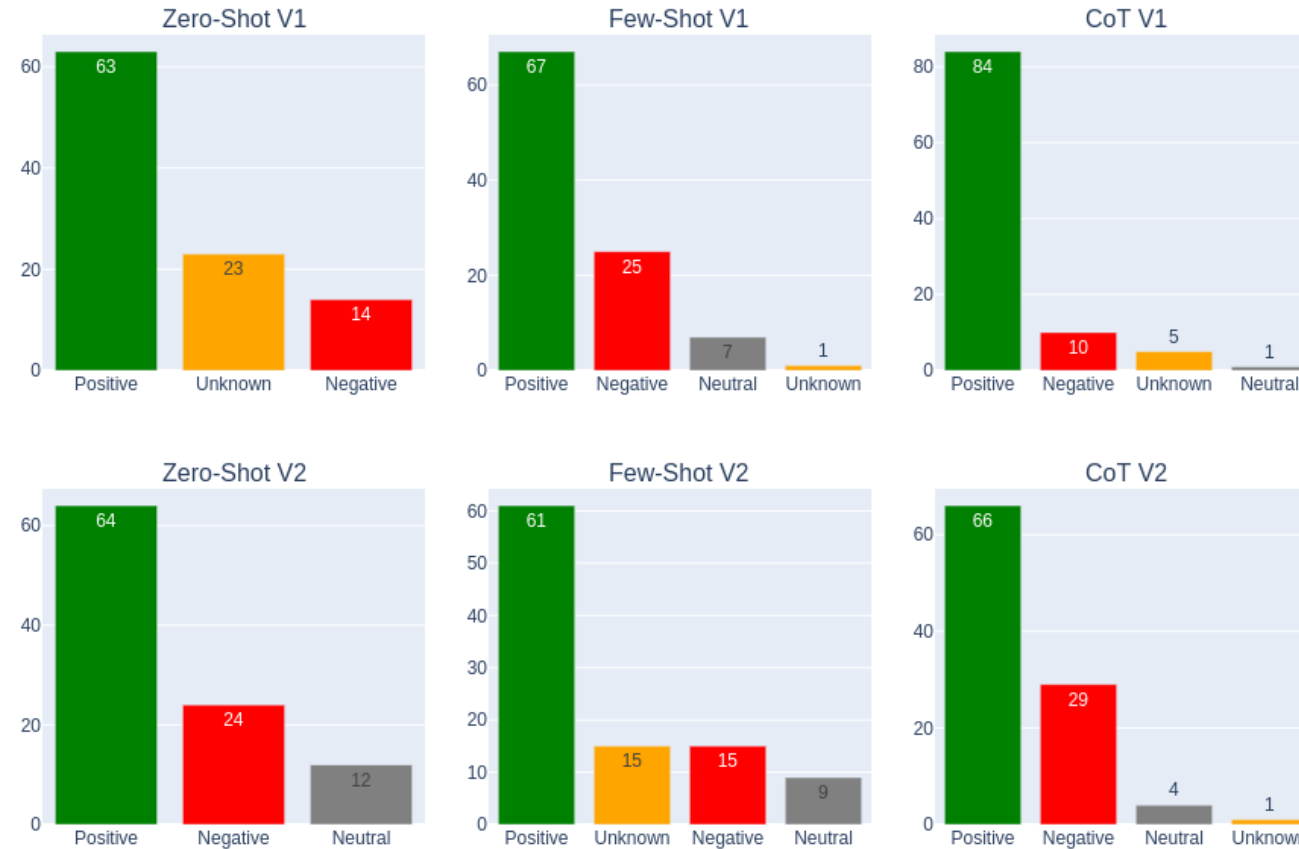
D: Classification report summary table

Metric	GPT-4o-mini	GPT-5-mini	Leading Model
Accuracy	87.0%	89.0%	GPT-5-mini
Precision (Class 1)	1.00	0.99	GPT-4o-mini
Recall (Class 0)	1.00	0.95	GPT-4o-mini
False Negatives	13	10	GPT-5-mini
False Positives	0	1	GPT-4o-mini

(Further ref. In final research notebook)

E: Sentiment Distrib. Dashboard (GPT-4o-mini)

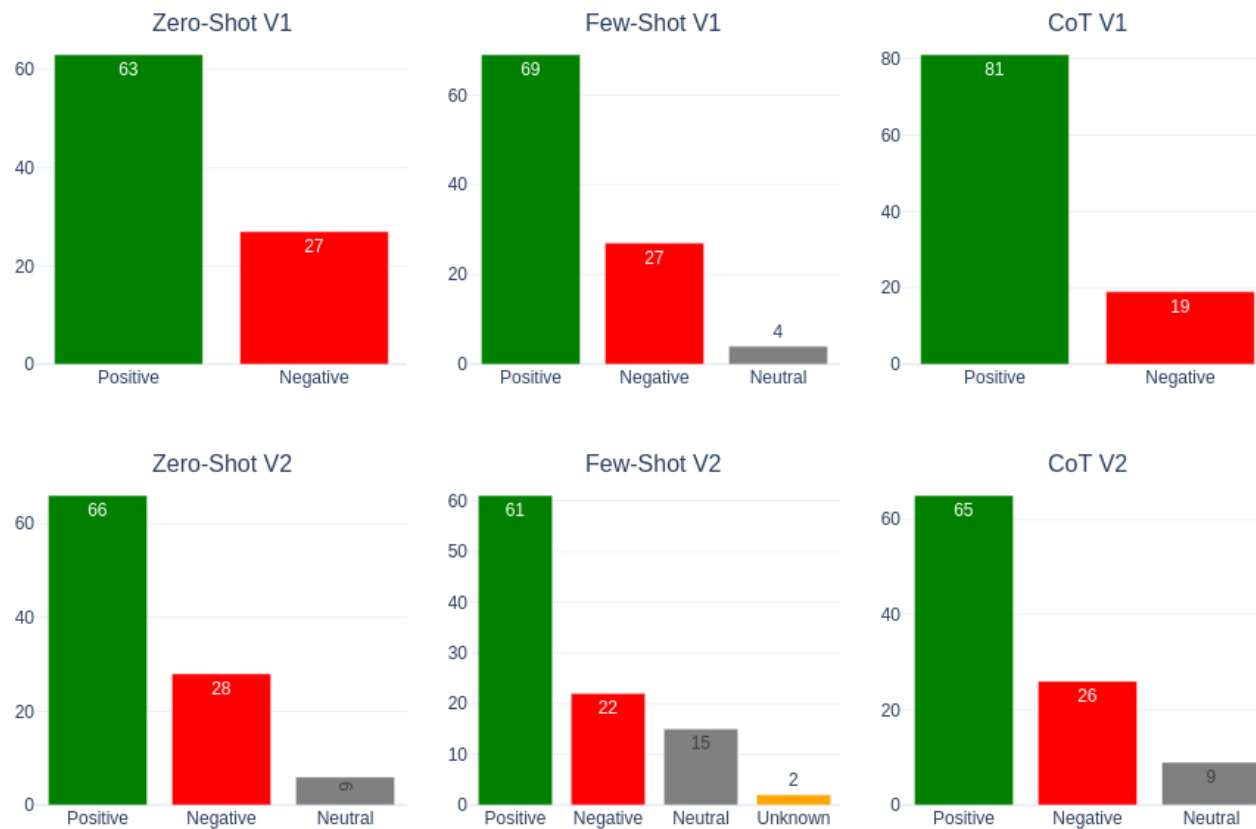
Sentiment Distribution Dashboard: GPT-4o-mini (V1 vs V2)



(Further ref. In final research notebook)

F: Sentiment Distrib. Dashboard (GPT-5-mini)

Sentiment Distribution Dashboard: GPT-5-mini



(Further ref. In final research notebook)